INCH-POUND

MIL-R-43161C
17 August 1990
SUPERSEDING
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MILITARY SPECIFICATION

ROPE, NYLON (SPUN YARN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and size of spun yarn nylon rope.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4020

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MILITARY

MIL-C-3131 - Cordage; Packaging Of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Government and contractor purchases. The requirements specified in 3.9, 3.10, and the related requirements of sections 4 and 5 apply only to rope purchased directly by the Government. All other requirements apply both to rope purchased by a contractor as a component for an end item and to rope purchased directly by the Government.

3.3 Standard sample. The rope shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.4 Material. The rope shall be fabricated from spun staple nylon yarn in a denier to meet the requirements of this specification.

3.5 Construction. The rope shall be of three strand construction to conform to the requirements specified herein. The lay of the rope shall be right-hand or "Z" lay. The length of ten turns of the finished rope shall be not less than 7 inches nor more than 8 inches when tested as specified in 4.4.4.

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3.6 Physical requirements.

3.6.1 Circumference. The circumference of the rope shall be 3/4 ± 1/8 inch when tested as specified in 4.4.4.

3.6.2 Length per pound. The length per pound of the rope shall be not less than 47 feet when tested as specified in 4.4.4.

3.6.3 Breaking strength. The rope shall have a minimum breaking strength of not less than 850 pounds, and the loss in breaking strength after heat aging of the rope shall not exceed 10 percent when tested as specified in 4.4.4.

3.6.4 Elongation. The elongation of the rope shall be no more than 35 percent when tested as specified in 4.4.4.

3.7 Color. Unless otherwise specified (see 6.2), the rope shall be dyed Olive Drab (see 6.4).

3.7.1 Colorfastness. The dyed and finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.2.5. When no standard sample has been established, the dyed rope shall show "good" fastness to weathering when tested as specified in 4.4.4.

3.7.2 Matching. The color of the finished rope shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.8 Extraneous material. No extraneous material shall be added for the purpose of weighting the rope. The extractable matter of the finished rope shall be no more than 4.0 percent when tested as specified in 4.4.4.

3.9 Identification. Each spool (reel) of rope shall have a ticket (identification tag) attached for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15-point paper stock and have a minimum tearing resistance in both directions (total) of 850 grams when tested as specified in UU-T-81. All ticket entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket shall contain the following information.

National stock number
Nomenclature
Specification number
Length
Size (circumference)

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Contract number and date
Contractor's name
Date of manufacture (month and year)

3.10 Put-up. Unless otherwise specified (see 6.2), the rope shall be put-up in nonreturnable spools (reels) in minimum 600-foot lengths. A plus tolerance of 10 percent is allowed on the length of any spool (reel) specified. The rope, when put-up in spools (reels), shall be in one continuous piece and so wound that each turn and layer is free from entanglement. The ends of all rope shall be cut off squarely and securely whipped or heat sealed to prevent fraying or untwisting.

3.11 Workmanship. The rope shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and shall be tested for the characteristics specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed in table I conform to the specified requirements.

TABLE I. Component tests

Characteristic	Requirement paragraph
Material	3.4

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of spools (reels). The sample unit shall be one spool (reel), but not less than 100 feet. The inspection level shall be I, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Any cut	101	
	Chafed or damaged	102	
	Kink	103	
	Broken or loose end	104	
	Bulged strand	105	
	Darting yarn 1/	106	
	Strand knot	107	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Cleanness	Spot or stain, clearly visible 2/	108	
Identification	Omitted, incorrect, illegible, or not as specified	109	
	Insecurely attached		201

1/ Darting yarns are internal yarns which project through the cover yarns of the strand.

2/ At normal inspection distance (approximately 3 feet).

4.4.3 End item length and winding examination. The rope shall be examined for the defects listed below. The lot size shall be expressed in units of spools (reels) of rope. The sample unit shall be one spool (reel) of rope. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number shall be 1.

Defects

Total length of rope on any spool is less than or greater than the length specified (see 3.10).
Length of rope on spool (reel) is less than or greater than the length marked on ticket (identification tag).
Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding, or otherwise affecting free unhampered unwinding of rope.
Any end frayed or not cut off squarely or neatly.
Any end not securely whipped or heat sealed.

4.4.4 End item testing. The end items shall be tested for the characteristics listed in table III. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table III shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in table III, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes specified in the applicable test methods. The lot size shall be expressed in units of spools (reels) of rope. The sample unit shall be one spool (reel) of rope. The sample size shall be in accordance with table IV. The lot shall be unacceptable if one or more sample units fail to meet any requirement. All test reports shall contain the individual values utilized in

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expressing the final results. Tests to determine compliance with document requirements may be made under prevailing atmospheric conditions. In case of dispute, tests shall be made upon material that has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE III. End item tests

Characteristic	Requirement reference	Test method	Number of determinations per individual sample unit	Results reported as
Number of strands	3.5	Visual	1	Pass or fail
Lay of rope	3.5	4050	-	-
Length per 10 turns	3.5	6000	-	-
Circumference	3.6.1	6003	-	-
Length per pound	3.6.2	6004	-	-
Breaking strength: Initial	3.6.3	6015	-	-
After heat aging	3.6.3	4.5.1	2	Average of 2 determinations to nearest 10 pounds calculated to 1 percent
Elongation	3.6.4	6015	-	-
Colorfastness: to weathering	3.7.1	5671 1/	-	-
Shade matching	3.7.2	3.7.2	1	Pass or fail
Finish (extraneous material)	3.8	2/	-	-
Extractable matter	3.8	2611	-	-

1/ The specimens shall be exposed for 40 standard fading hours.

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2/ Unless otherwise specified, a supplier's certificate of compliance shall be furnished and will be acceptable for the stated requirement.

TABLE IV. Sample size

Lot size spools (reels)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.5 Packaging inspection. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-C-3131, except that the inspection level shall be S-2, and the AQL shall be 2.5.

4.5 Methods of inspection.

4.5.1 Breaking strength after aging. Two breaking strength specimens, prepared as specified in Method 6015 of FED-STD-191, shall be heated for 5 days in a convection air oven at 175° ± 2°F. After aging, the specimens shall be allowed to reach equilibrium under standard conditions before being tested for strength. Breaking strength shall be determined in accordance with Method 6015. The percent change in strength shall be calculated based on the average strengths initially and after heat aging.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The rope, put-up as specified in 3.10, shall be preserved in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The rope shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-C-3131.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The rope is intended to be used for lacing pontoon floats.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Put-up, if other than specified (see 3.9).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Color, if other than Olive Drab (see 3.7).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Lacing
Pontoon
Spools

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - IS

User activity:

Army - ME
Navy - SH

Preparing activity:

Army - GL

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